

Spoken Tutorial on Understanding APIs

Module	An introduction to Building
Episode	2. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is. 2. Explain the role of prompts and API keys. 3. Identify how applications send requests and receive responses.
Approx. Duration	3-4 min
Outline	• What an API is • API as a connector • Prompts and API keys • Application interaction with APIs
Meta Tags	API, Application Programming Interface, prompts, API keys, authentication, access control, AI, requests, responses, connector, Python basics, tutorial
Pre-requisite Tutorial	Episode 1: Python basics for API

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Understanding APIs .
Learning Objectives Slide	In this tutorial, you will learn to, • Define what an API is. • Explain prompts and API keys. • Identify how applications interact with APIs.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have watched Episode 1: Python basics for API .
Restaurant analogy: waiter connecting customer and kitchen	Have you ever wondered how different apps talk to each other? They use something called an API . API stands for Application Programming Interface . It is a set of rules that lets software programs communicate. Think of a restaurant. You, the customer, order from the waiter. The waiter takes your order to the kitchen. The kitchen prepares food and gives it back to the waiter. The waiter then brings your food to you. Here, the waiter is the API . It connects you to the kitchen without you knowing details. Similarly, an API connects different software parts.

Person at laptop, illustration of prompt going into API and API key unlocking access	With that understanding, let's look at how we use APIs. When you use an AI tool, you give it a prompt. A prompt is simply the input data you send to the API. For example, asking ChatGPT a question is a prompt. But how does the API know who you are? This is where API keys come in. An API key is a unique code. It acts like a digital ID card or a password. You send your API key with your prompt. This key authenticates your request. It ensures only authorized users can access the API.
Screen close-up: user typing 'Imagine an application wants to use an AI model...'	Now, let's see a simple example of sending a request. Imagine an application wants to use an AI model. It might try to send a prompt directly like this. See how the request is very basic. There is no clear structure for the data. Also, there is no API key included. This method is unoptimized and unsecured. It is like shouting your order in a restaurant. The kitchen might not understand you. Or, it might not know who placed the order.
Screen close-up: user typing 'Here, the application sends a structured request...'	Now, let's look at an improved way to send a request. Here, the application sends a structured request. Notice the data is organized, perhaps in JSON format. It clearly states the prompt and other parameters. Most importantly, an API key is included for authentication. This is like giving a clear, written order to the waiter. You also show your membership card for access. The API can now process this request efficiently. It knows who you are and what you want.
Side-by-side: vague prompt output vs improved prompt output	Let us compare these two request methods. On one side, we have the vague, unstructured request. On the other, the improved, structured request. See the difference in clarity and security? The improved method uses a clear data format. It also includes the essential API key. This ensures your request is understood and authorized. It also protects your access to the API. Using structured requests and API keys is vital. It makes your application's interaction with APIs reliable.
Summary Slide	Let us summarize what we learned. We defined what an API is, like a digital waiter. We saw how prompts are inputs to APIs. We understood that API keys control access. Finally, we compared different ways to send requests.

Assignment Slide	Now as an assignment, Think about an app you use daily. How might it use an API to function? Describe the prompts and potential data flow. Compare your thoughts with a friend.
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